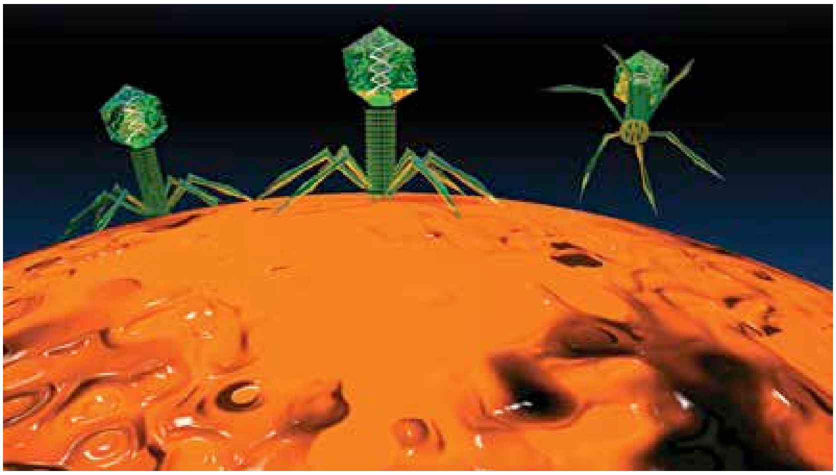
Aussie wind and solar farm
Australia has officially launched their first hybrid wind and solar farm: the Gullen Solar Farm. **http://dgit.in/ausfarms**

Pro gun russian bots
Reports claim that the Russian government flooded Twitter with pro-gun bots soon after the Parkland shooting incedent. **http://dgit.in/progbots**

Combating drug resistant bacteria with viruses



Bacteria are evolving resistances faster than the rate at which we can develop new antibiotics. An emerging treatment option is the use of viruses known as bacteriophages.

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Indiscriminate and uncontrolled use of antibiotics has led to a situation where the bacteria are evolving immunity to various antibiotics at an alarming rate. Bacteria can pass on the genetic material to other bacteria, making them drug resistant as well. Early in 2017, the UN identified twelve particularly harmful strains of bacteria, that had developed resistance to the best treatment options available. These bacteria are responsible for life threatening diseases including bloodstream infections and pneumonia.

What makes these bacteria particularly harmful, is that they thrive and spread in an environment often visited by humans when their natural defenses are already weak – in hospitals. The bacteria can spread through the devices used to treat patients, including ventilators and catheters.

An emerging treatment option is the use of viruses to attack the harmful bacteria. Bacteriophages are viruses that infect bacteria, then reproduce within them.

Shortly before the advent of antibiotics in the 1940s, there was actually some research into the use of bacteriophages for treating infections. The interest in the research ceased as the antibiotics proved to be a potent treatment, at least for a time. Now, there is a rekindled interest in the use of bacteriophages.

Bacteriophages are highly specific to particular species or families of bacteria. They are harmless to humans, and also to other beneficial bacteria within humans. Antibiotics tend to take down beneficial bacteria as well. Thus, treatment options can be designed to target only the harmful bacteria. If there is more than one strain causing an infection, a combination of bacteriophages, or bacteriophage cocktails can be administered.

Just as bacteria evolved resistance to antibiotics, they can evolve resistance to phages as well. This evolution can happen during the treatment itself, before the pathogen is eliminated from the host body. The advantage of using viruses is that as living organisms, they have the capacity to evolve as well, in order to counter the resistance of the bacteria.

There are a handful of companies working on another approach – bioengineered bacteriophages that are particularly effective at killing the target bacteria.

Within bacteria, are DNA sequences that are collectively known as CRISPR, which stands for Clustered Regularly Interspaced Short Palindromic Repeats. CRISPR is a natural defense mechanism for the bacteria, and contain portions of DNA from the bacteriophages, that help the bacteria identify and eliminate the viruses. Bioengineered bacteriophages infect the target bacteria, and modify the CRISPR system, prompting the bacteria to use its natural defenses to destroy itself. One approach for these modified bacteriophages involves completely removing the ability for the bacteriophage to replicate at all. The bacterial immune system is essentially repurposed to cause the pathogen to self destruct.

Human trials are expected to begin in Europe within the next two years. As the capacity for antibiotics to tackle bacteria reduces, there is expected to be an increasing reliance on bacteriophages for treating infections.